SOFTWARE ANALYSIS DOCUMENTATION
USE CASES

RENTAMOVIL

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ANÁLISIS Y DESARROLLO DE SOFTWARE 3145556

SERVICIO NACIONAL DE APRENDIZAJE

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1. Introduction

This document presents RentaMovil's use cases: what each actor can do and under what conditions. It is the first of the four dynamic views of the software analysis; the other three —sequence, activity and state diagrams— are in separate documents inside this same folder.

Each use case is presented as an individual card: its own diagram, actor, description, precondition, postcondition and the SRS functional requirement it implements.

2. System actors

Actor	Description
Visitor	A person without a session; can only browse the catalog and sign up
Customer	Authenticated user with the `CLIENT` role; manages their own reservations, payments and profile
Administrator	Authenticated user with the `ADMIN` role; manages the fleet, reservations and payments for the whole company
Super Admin	Authenticated user with the `SUPER_ADMIN` role; everything the Administrator can do, plus managing administrative users and bank accounts

> *An Administrator or Super Admin can also hold their own reservations, using the same account — the role determines what they can administer, not what they can use.*

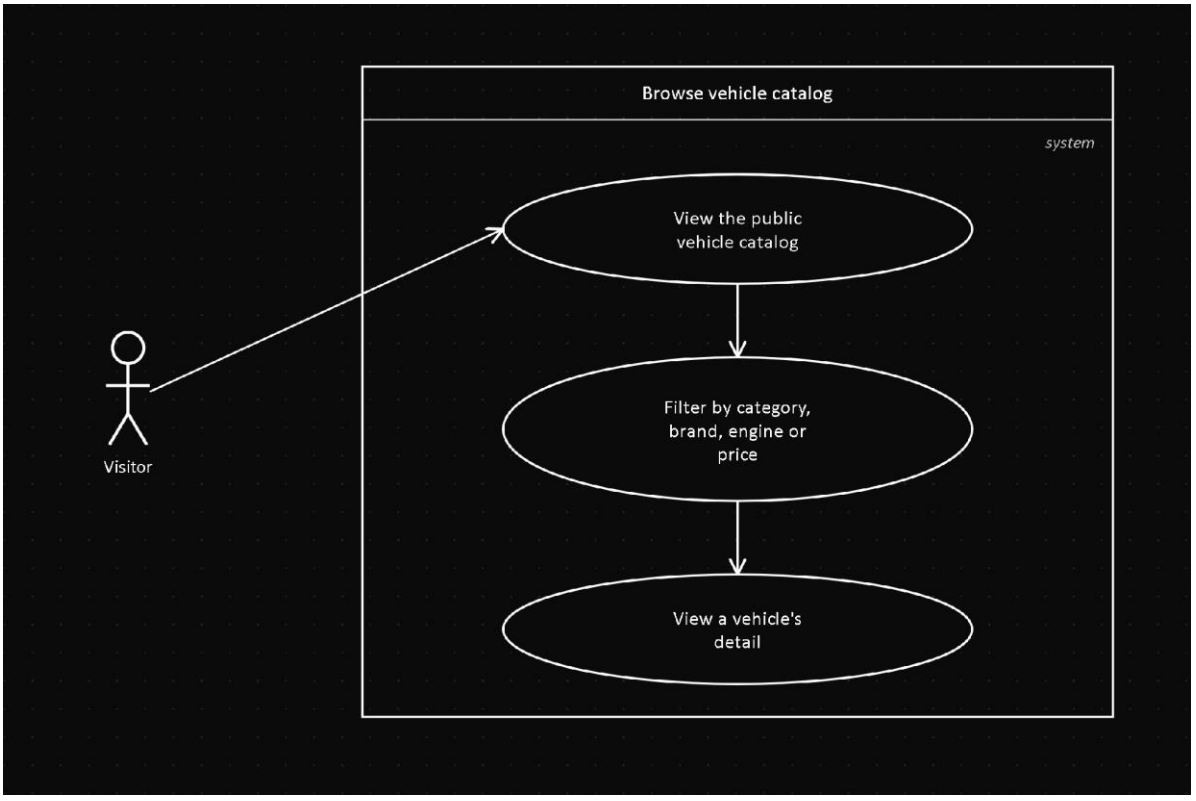
3. Use case catalog

ID	Use case	Actor
UC01	Browse vehicle catalog	Visitor
UC02	Sign up	Visitor
UC03	Log in	Customer
UC04	Recover password	Customer
UC05	Manage my profile	Customer
UC06	Create a reservation	Customer
UC07	View and modify my reservations	Customer
UC08	Cancel a reservation	Customer
UC09	Pay for a reservation	Customer
UC10	View my invoices	Customer
UC11	View my notifications	Customer

ID	Use case	Actor
UC12	Manage vehicles	Administrator
UC13	Manage branches	Administrator
UC14	Manage fleet maintenance	Administrator
UC15	Register pickup and return	Administrator
UC16	Review payment receipts	Administrator
UC17	Manage company reservations	Administrator
UC18	Track vehicle location	Administrator
UC19	Manage insurance plans	Administrator
UC20	Manage administrative users	Super Admin
UC21	Manage bank accounts	Super Admin

4. Use case cards

UC01 — Browse vehicle catalog

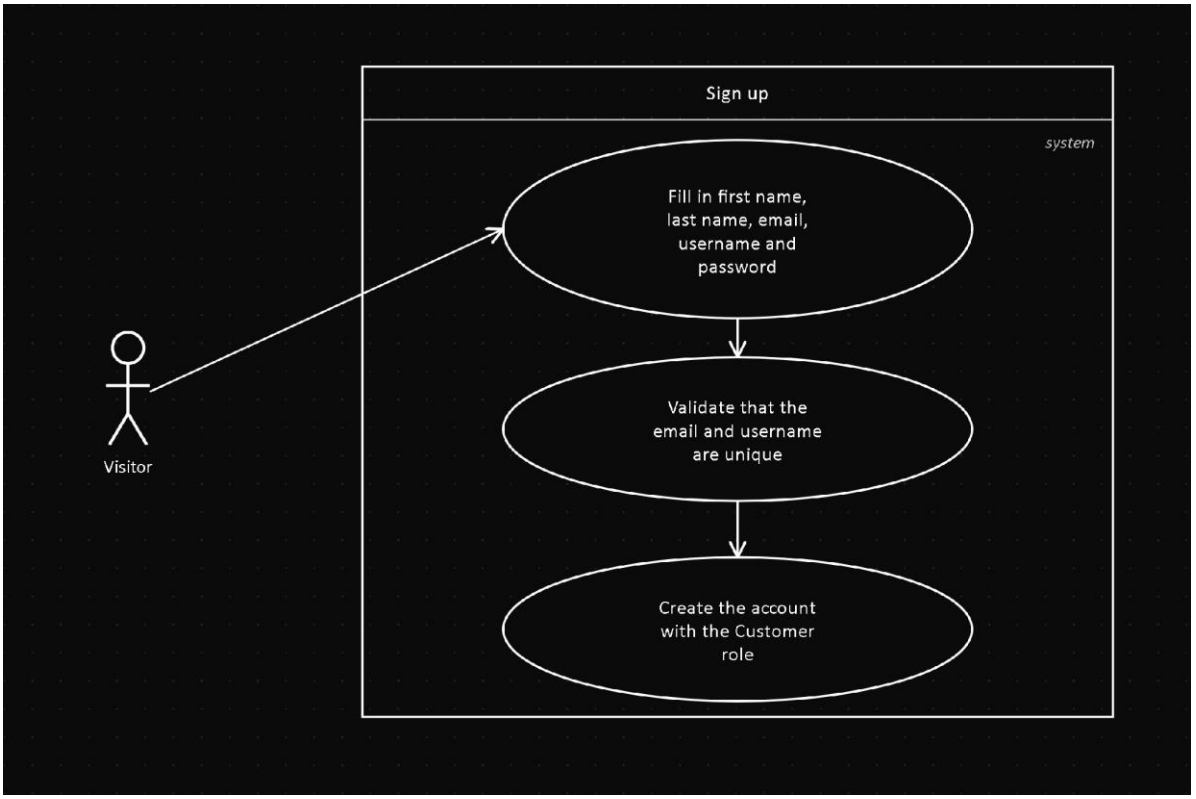


Field	Value
Actor	Visitor
Related FR	FR2.1
Precondition	None — no session required

Field	Value
Postcondition	The list of `AVAILABLE` vehicles is shown, with optional filters

Description: any visitor can view and filter (category, brand, engine, price, dates) the public vehicle catalog, without needing to register.

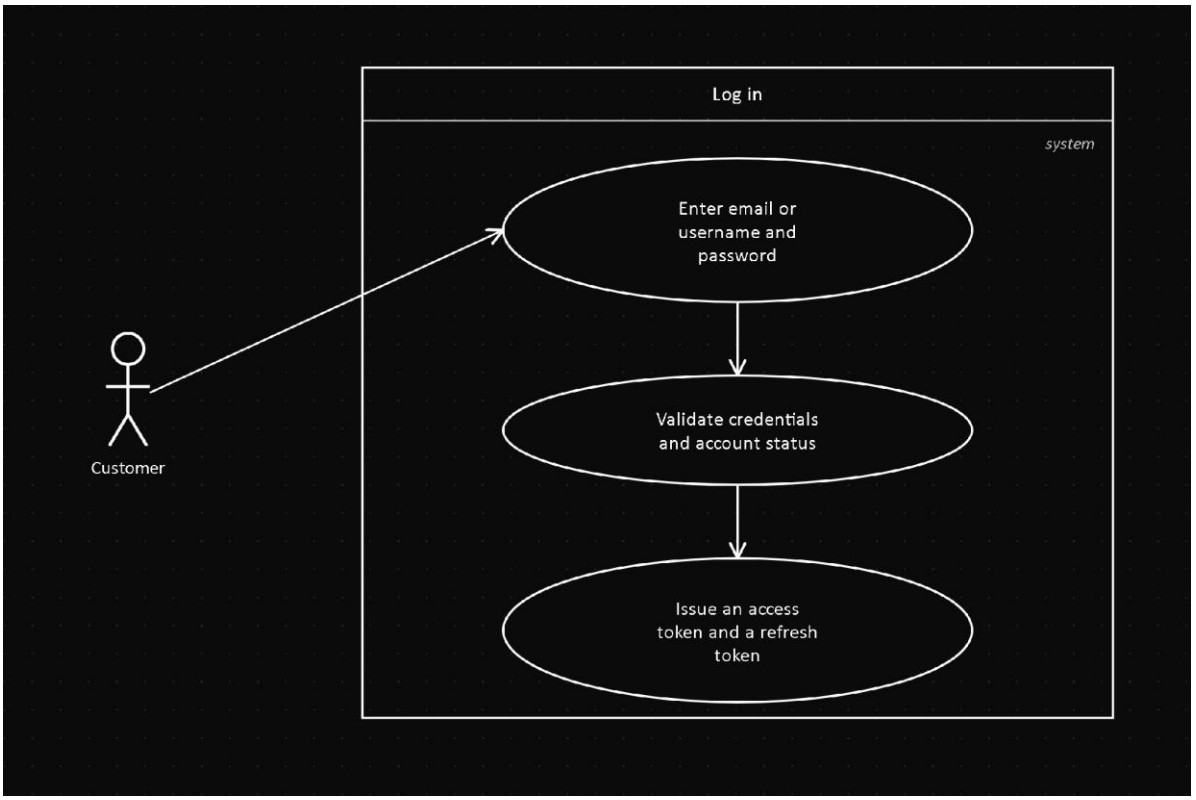
UC02 — Sign up



Field	Value
Actor	Visitor
Related FR	FR1.1
Precondition	The visitor has no account in the system
Postcondition	A Person and a User are created with the `CLIENT` role

Description: the visitor fills in first name, last name, email, username, password and, optionally, a phone number, to create their Customer account.

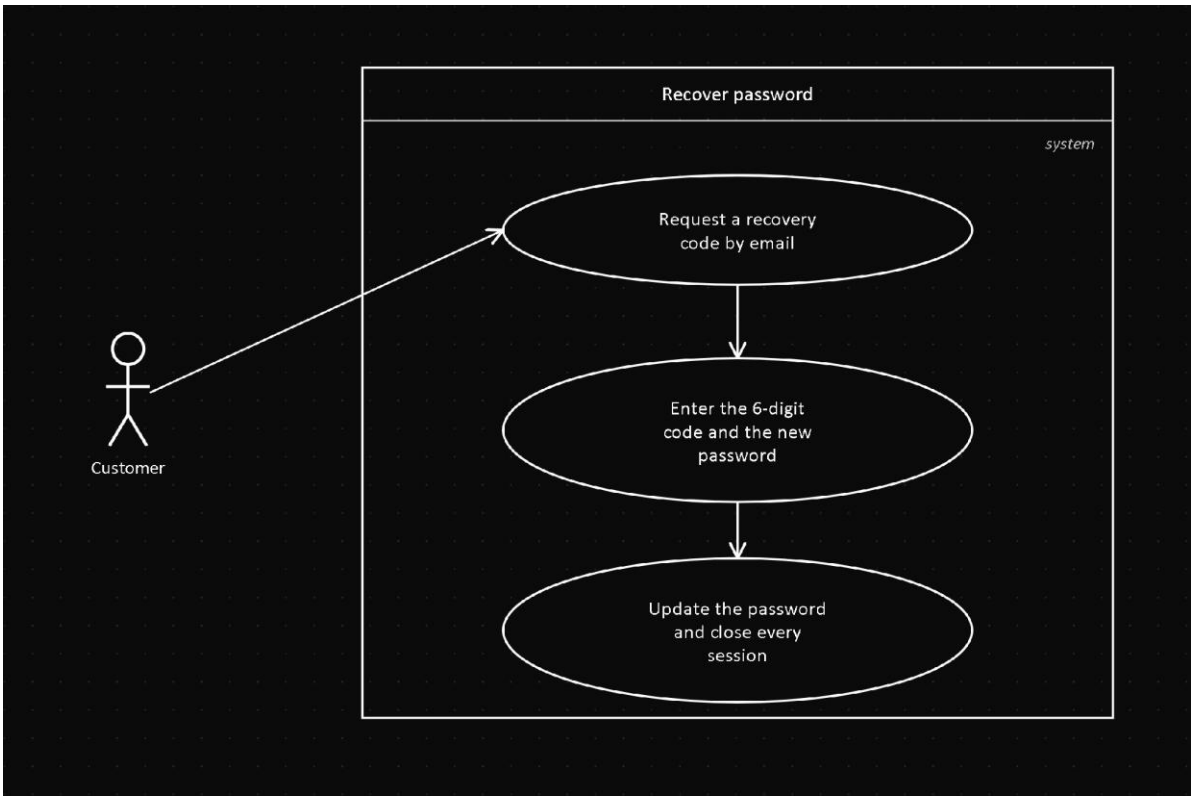
UC03 — Log in



Field	Value
Actor	Customer
Related FR	FR1.2
Precondition	The user has an account with `status = ACTIVE`
Postcondition	An access token (30 min) and a refresh token (7 days) are issued

Description: the user authenticates with their email or username and their password.

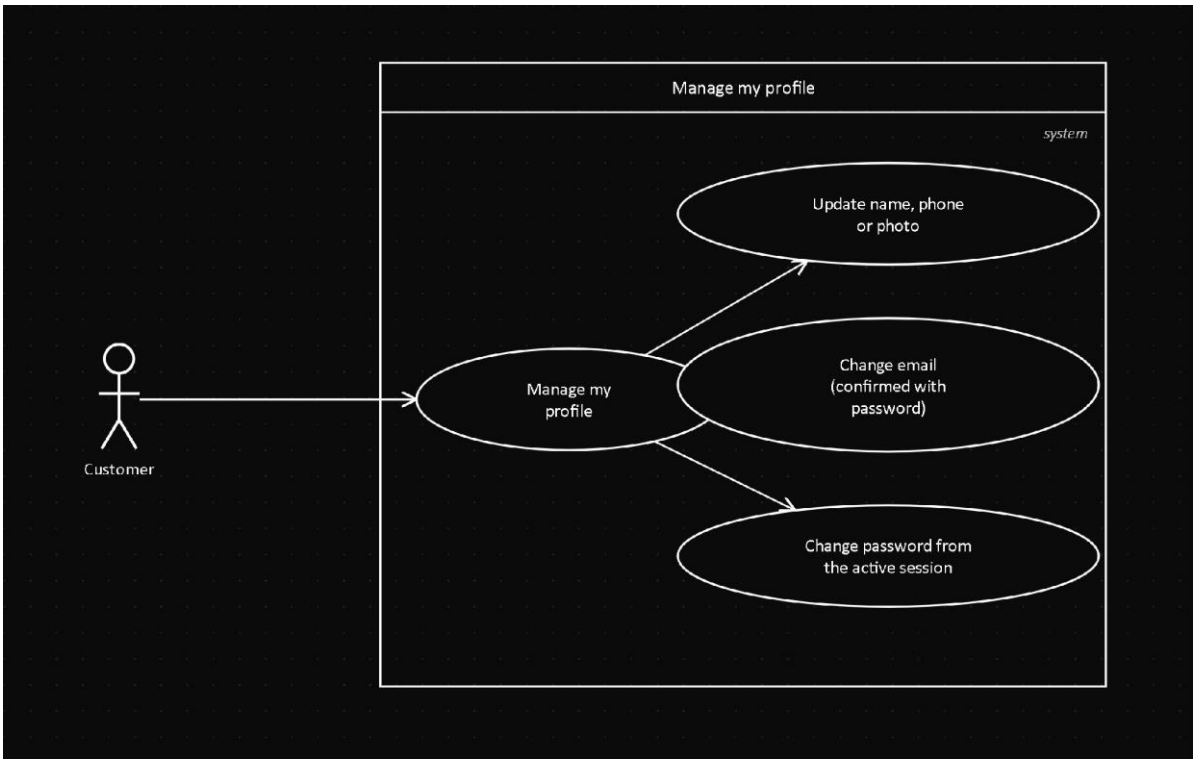
UC04 — Recover password



Field	Value
Actor	Customer
Related FR	FR1.3
Precondition	The user forgot their password and has a registered account
Postcondition	The password is updated and all of their sessions are closed

Description: the user requests a recovery code by email (6 digits, valid for 15 minutes) and uses it together with a new password.

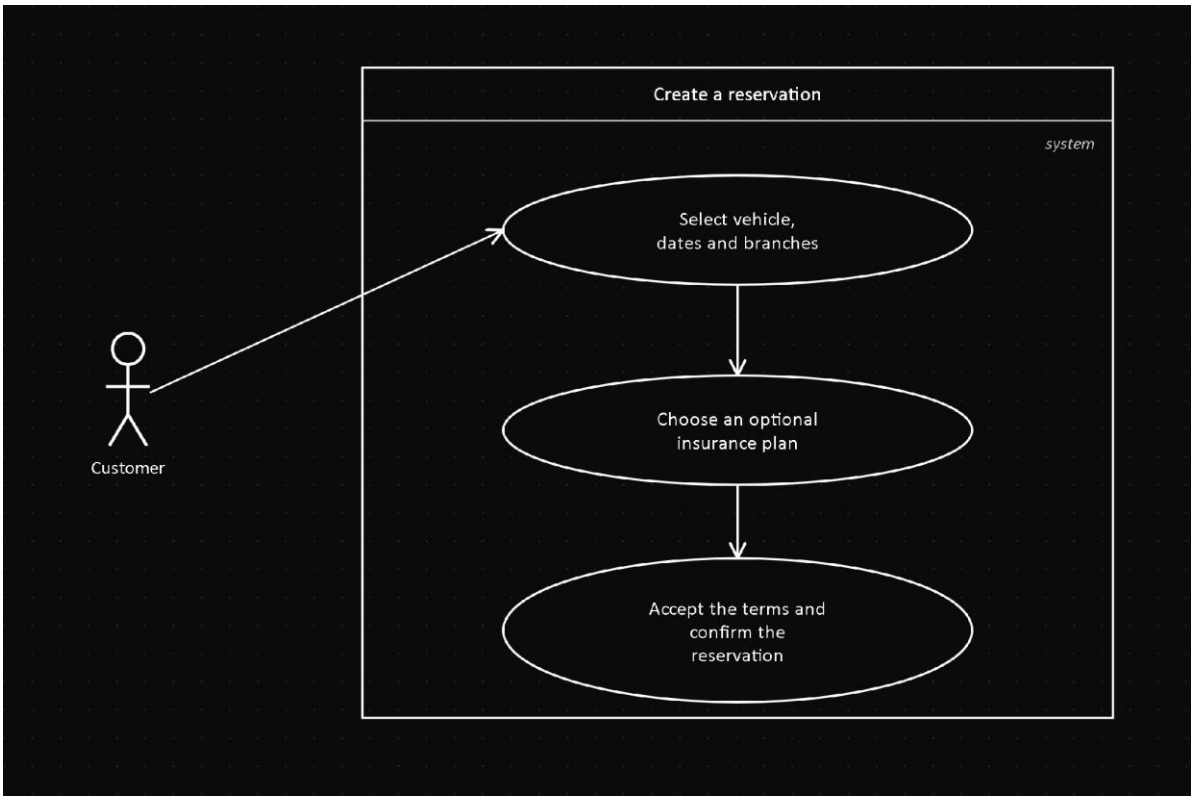
UC05 — Manage my profile



Field	Value
Actor	Customer
Related FR	FR1.4, FR1.5
Precondition	The Customer is authenticated
Postcondition	The profile data is updated

Description: the Customer updates their name, phone, photo or email (the latter, confirmed with their current password), and can change their password from the active session.

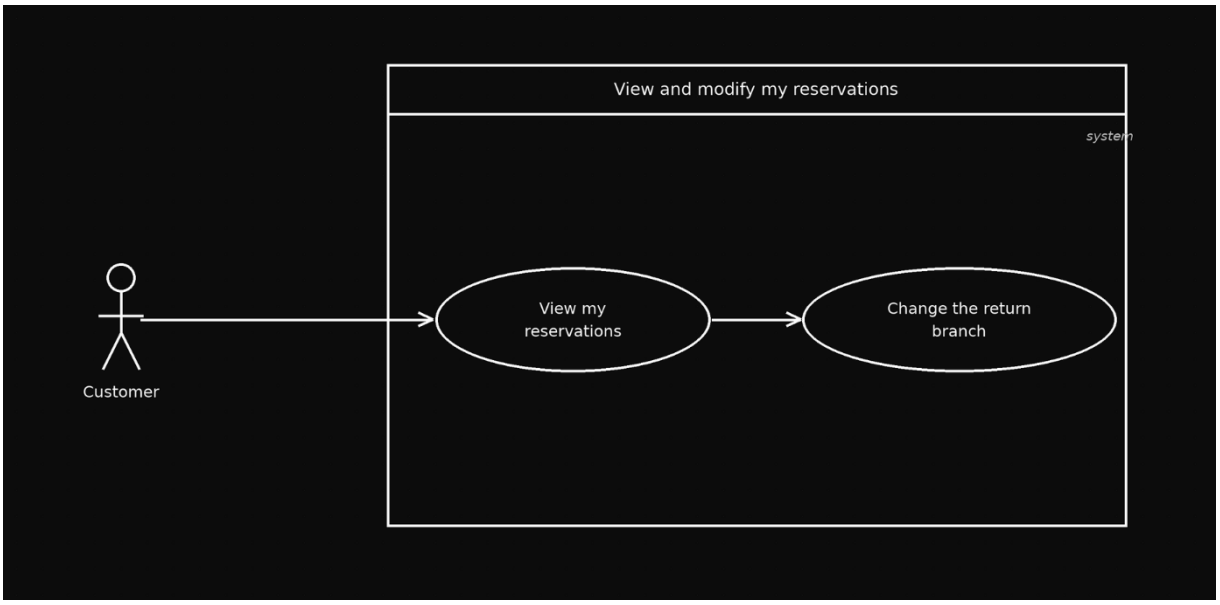
UC06 — Create a reservation



Field	Value
Actor	Customer
Related FR	FR3.1, FR2.2
Precondition	Customer authenticated; the chosen vehicle is available for the requested dates
Postcondition	A `Reservation` is created in `PENDING_PAYMENT` status

Description: the Customer selects a vehicle, dates, pickup/return branches and, optionally, an insurance plan, and accepts the terms and conditions.

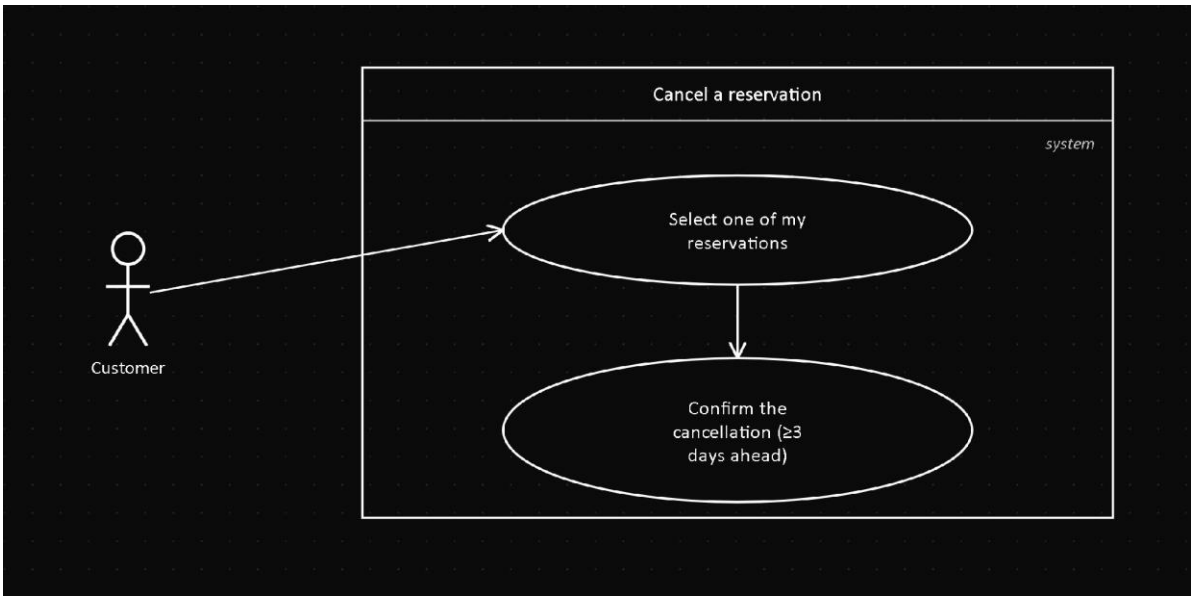
UC07 — View and modify my reservations



Field	Value
Actor	Customer
Related FR	FR3.2
Precondition	The Customer has at least one reservation
Postcondition	The return branch is updated

Description: the Customer views their reservations and can change the return branch, always with at least 3 days of notice.

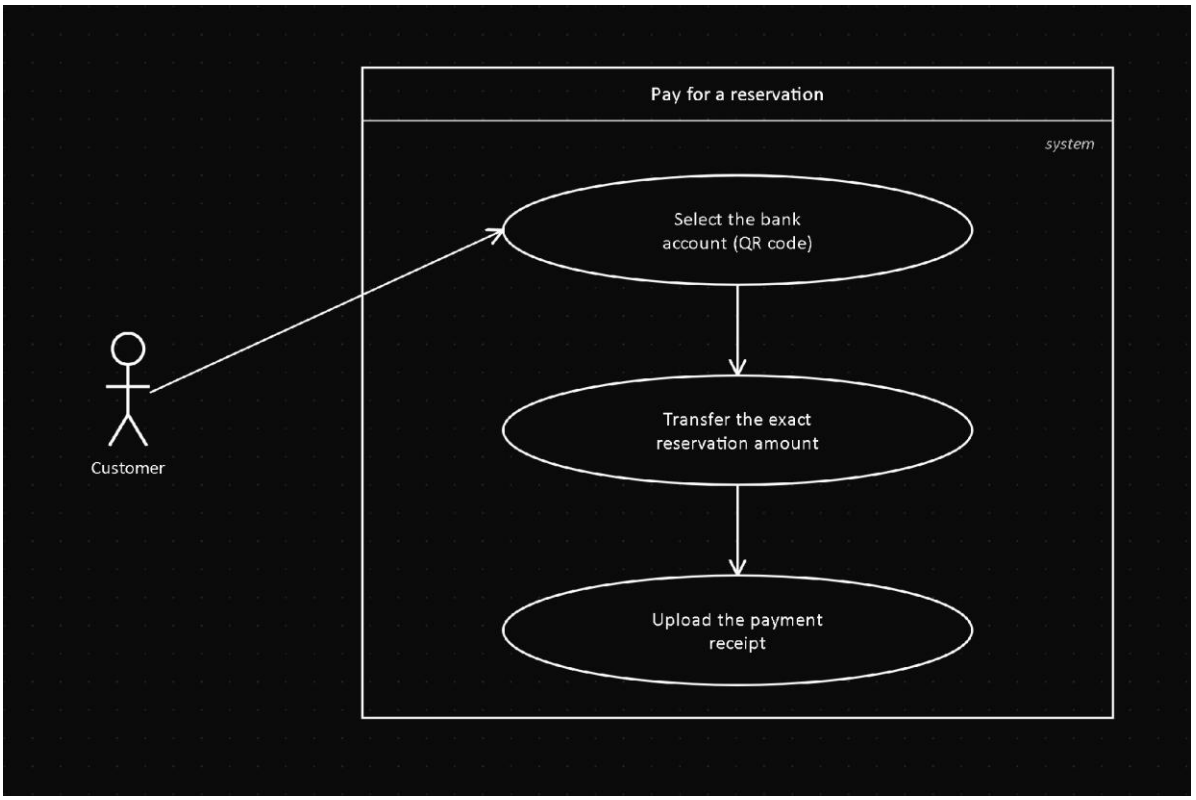
UC08 — Cancel a reservation



Field	Value
Actor	Customer
Related FR	FR3.3
Precondition	There are 3 or more days left before `start_date`
Postcondition	`Reservation.status = CANCELLED`, with no charge

Description: the Customer cancels their own reservation with no penalty, as long as there are at least 3 days left before the pickup date.

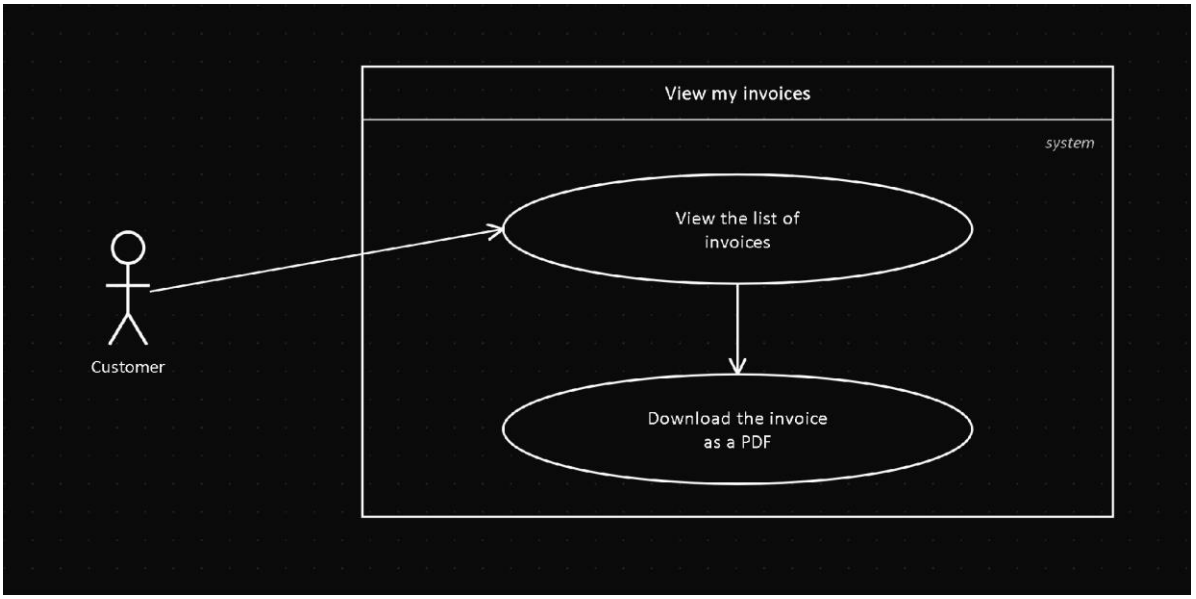
UC09 — Pay for a reservation



Field	Value
Actor	Customer
Related FR	FR5.1
Precondition	Reservation in `PENDING_PAYMENT`
Postcondition	A `Payment` is created in `PENDING_REVIEW`; the reservation moves to `PENDING_REVIEW`

Description: the Customer transfers to the bank account shown as a QR code and uploads the payment receipt for an Administrator to review.

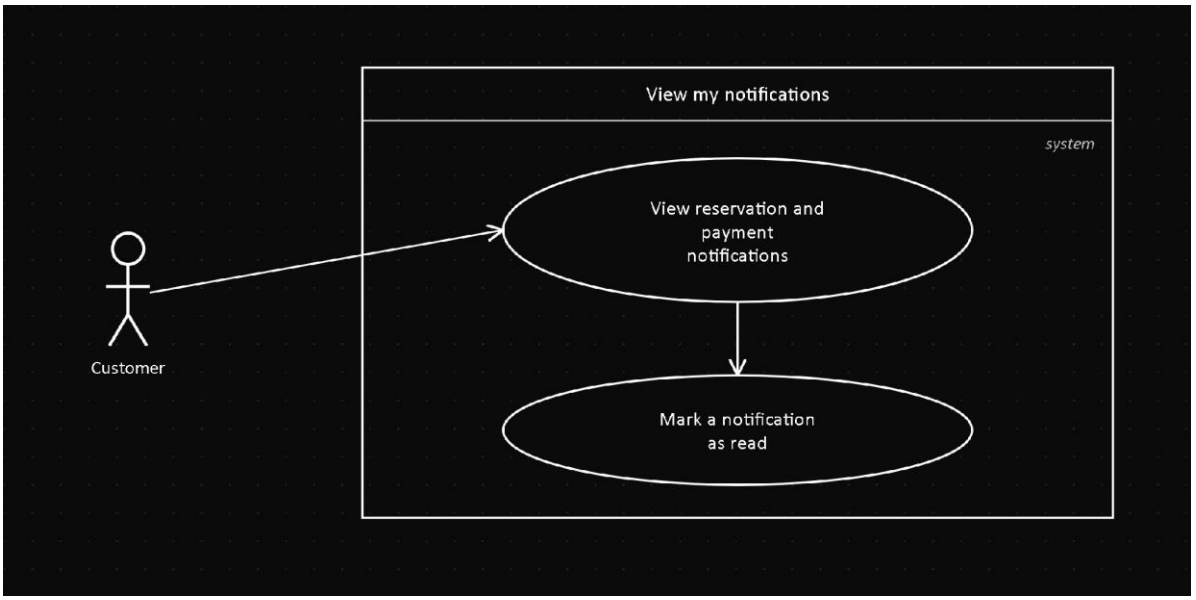
UC10 — View my invoices



Field	Value
Actor	Customer
Related FR	FR5.2
Precondition	The Customer has at least one approved payment
Postcondition	The invoice PDF is shown or downloaded

Description: the Customer views and downloads the invoices automatically generated after an approved payment.

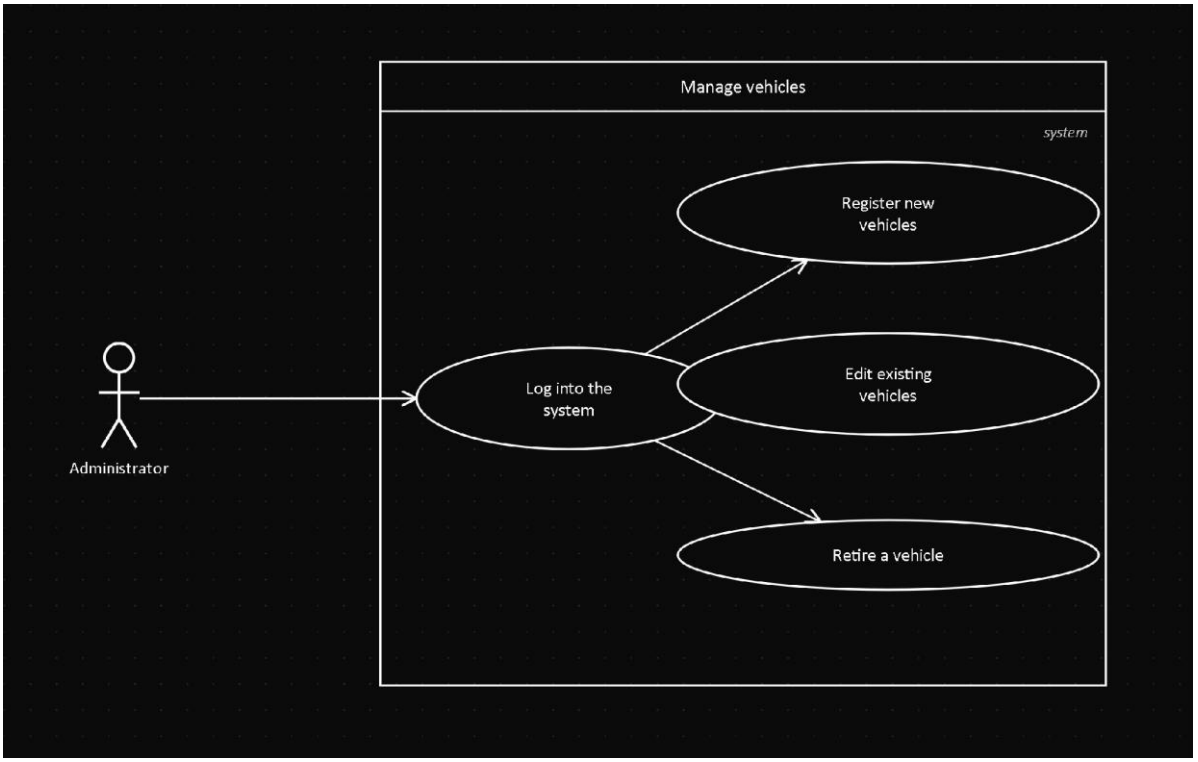
UC11 — View my notifications



Field	Value
Actor	Customer
Related FR	FR7.1
Precondition	The Customer is authenticated
Postcondition	The viewed notifications can be marked as read

Description: the Customer sees, inside the application, the notifications generated by events on their reservations and payments.

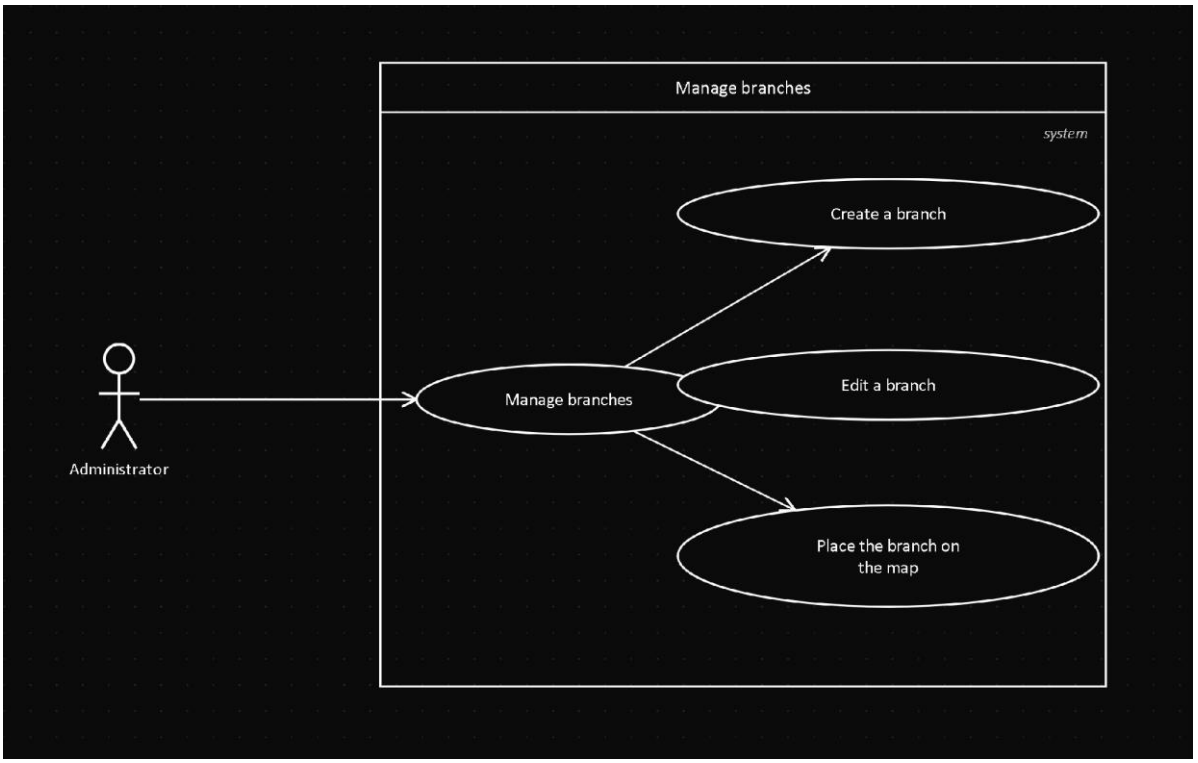
UC12 — Manage vehicles



Field	Value
Actor	Administrator
Related FR	FR8.2
Precondition	The Administrator is authenticated
Postcondition	The vehicle catalog is updated

Description: the Administrator registers, edits or removes vehicles, and views the full inventory filtered by status.

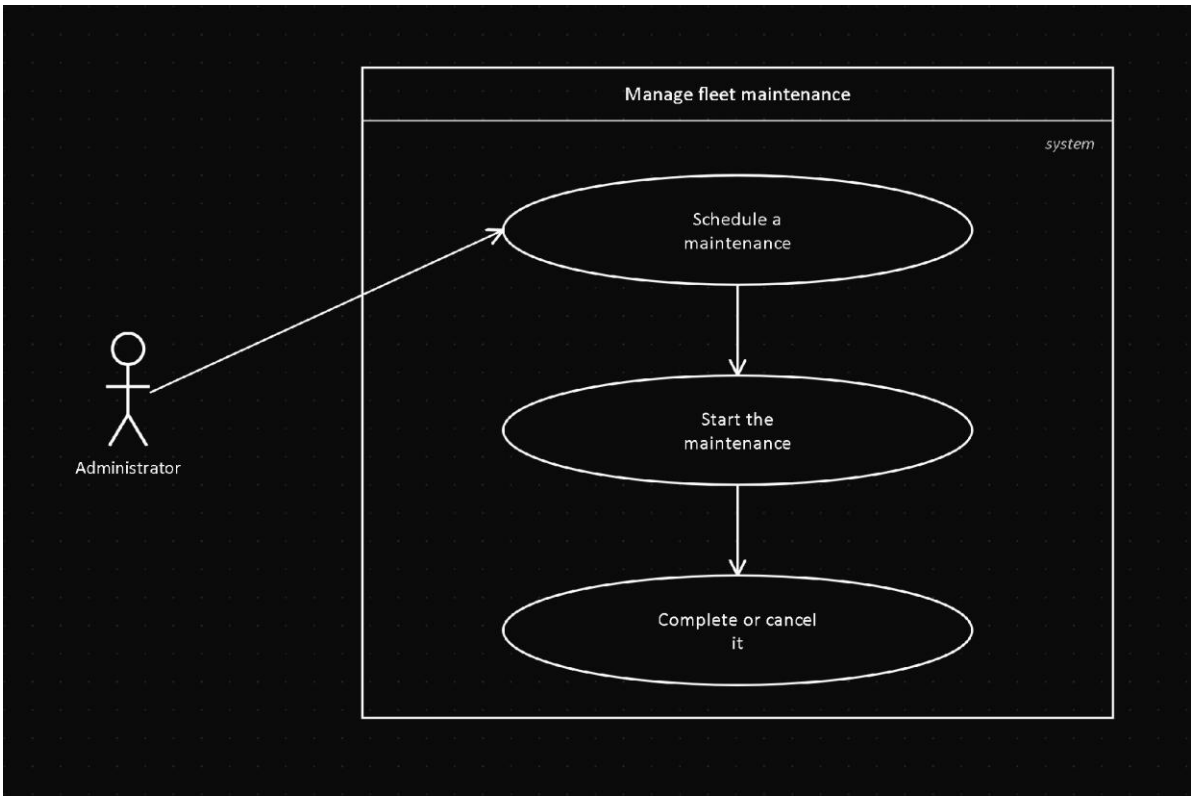
UC13 — Manage branches



Field	Value
Actor	Administrator
Related FR	FR8.4
Precondition	The Administrator is authenticated
Postcondition	The branch is created, edited or removed

Description: the Administrator creates, edits or removes branches, and can place them on a map.

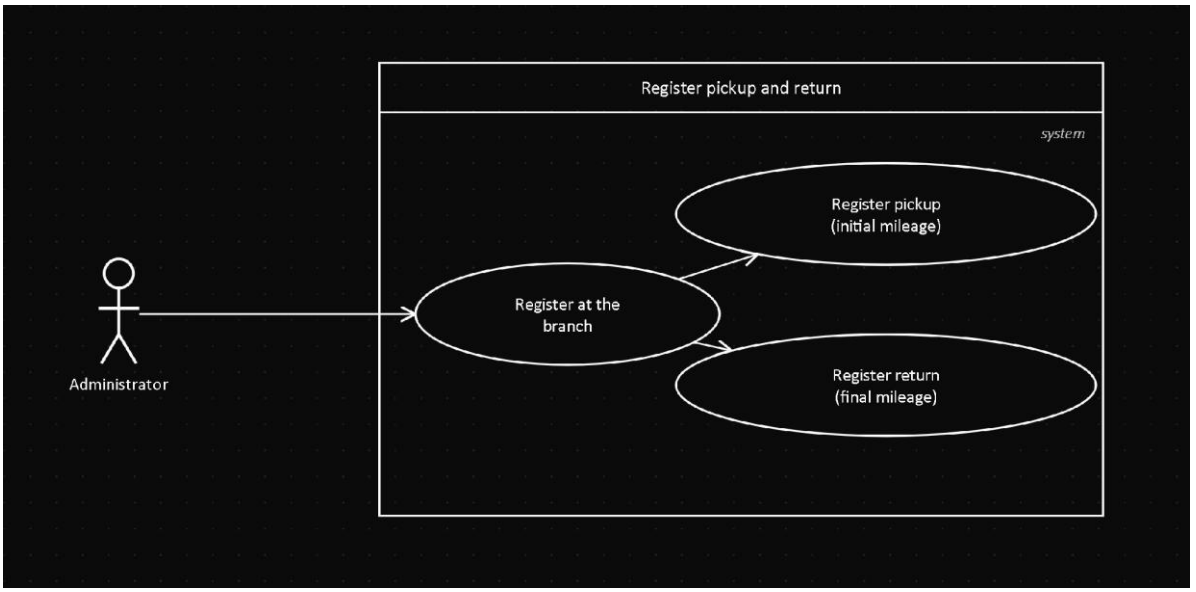
UC14 — Manage fleet maintenance



Field	Value
Actor	Administrator
Related FR	FR8.3, FR8.7
Precondition	The vehicle is not `RETIRED`
Postcondition	The maintenance is scheduled, started, completed or cancelled

Description: the Administrator schedules, starts, completes or cancels a vehicle's maintenance, and views its history.

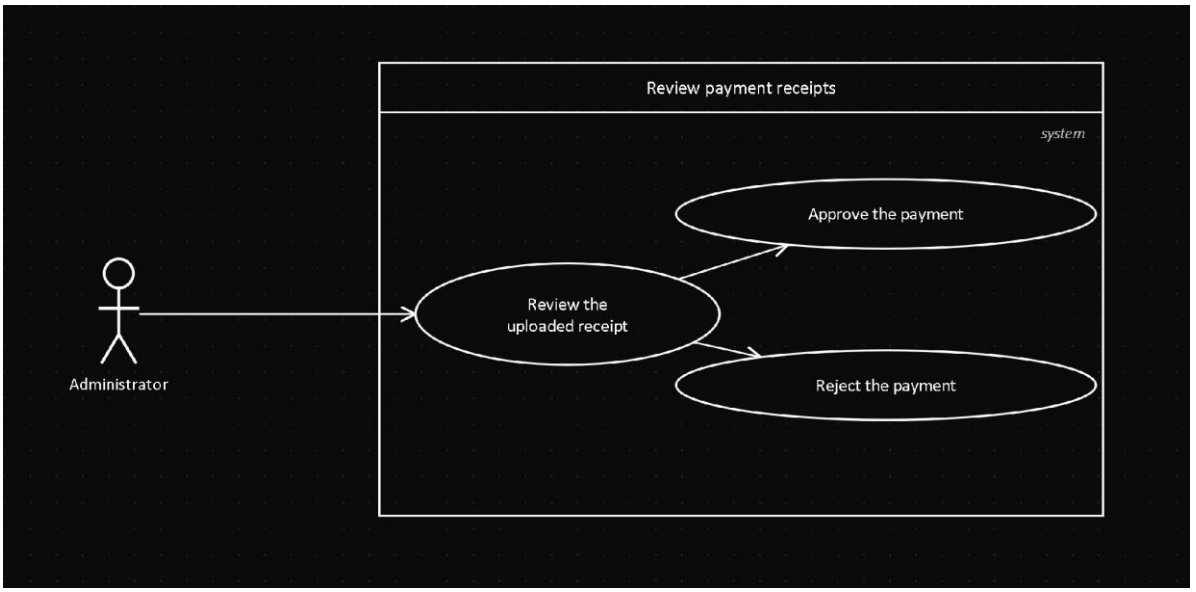
UC15 — Register pickup and return



Field	Value
Actor	Administrator
Related FR	FR4.1, FR4.2
Precondition	Reservation in `CONFIRMED` (for pickup)
Postcondition	A `Rental` is created or completed; the vehicle's status changes

Description: the Administrator registers, at the branch, the physical delivery and return of the vehicle, with the initial and final mileage.

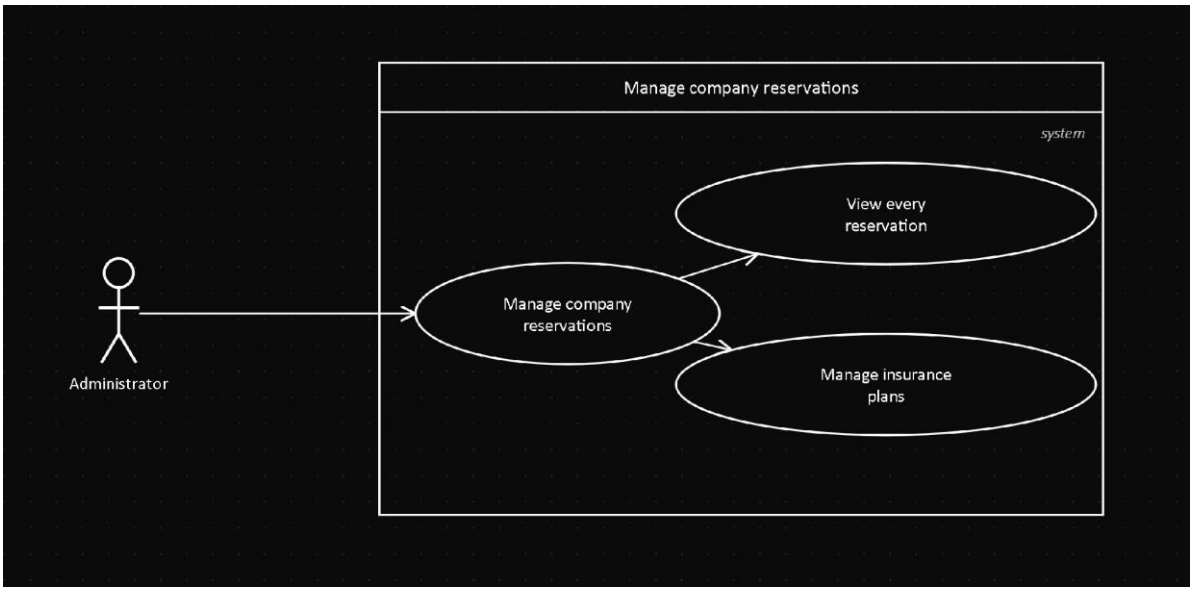
UC16 — Review payment receipts



Field	Value
Actor	Administrator
Related FR	FR5.1, FR8.5
Precondition	There is a `Payment` in `PENDING_REVIEW`
Postcondition	The payment ends up `APPROVED` or `REJECTED`

Description: the Administrator reviews the receipt uploaded by the Customer and decides to approve or reject it.

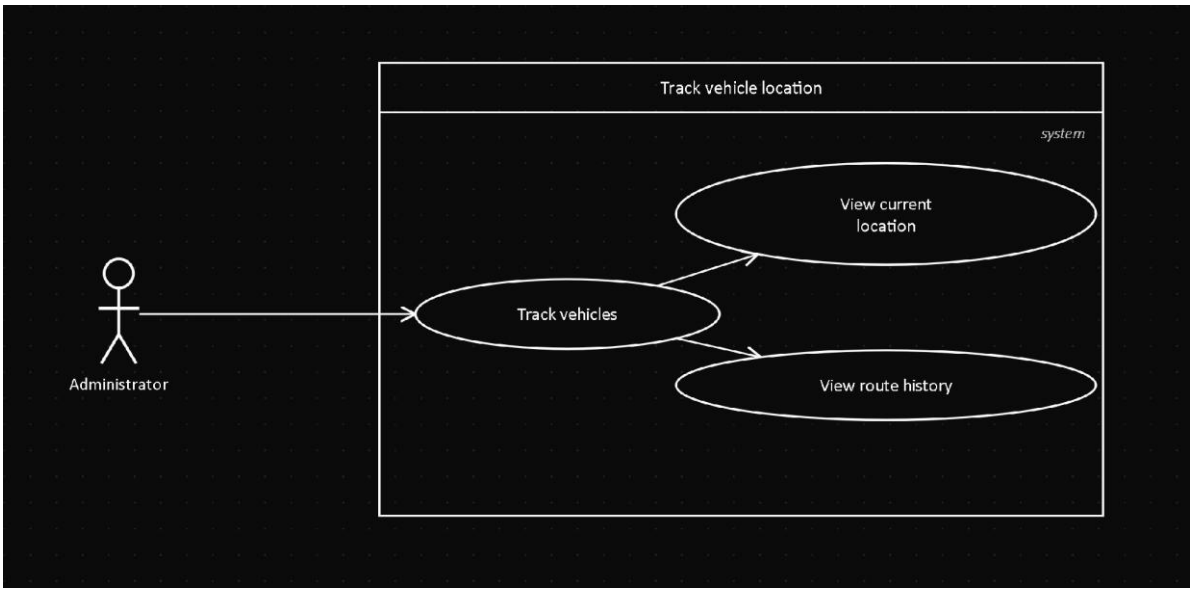
UC17 — Manage company reservations



Field	Value
Actor	Administrator
Related FR	FR8.5
Precondition	The Administrator is authenticated
Postcondition	The full list of reservations is shown, with its payment information

Description: the Administrator views every reservation in the system, not only their own, and manages the available insurance plans.

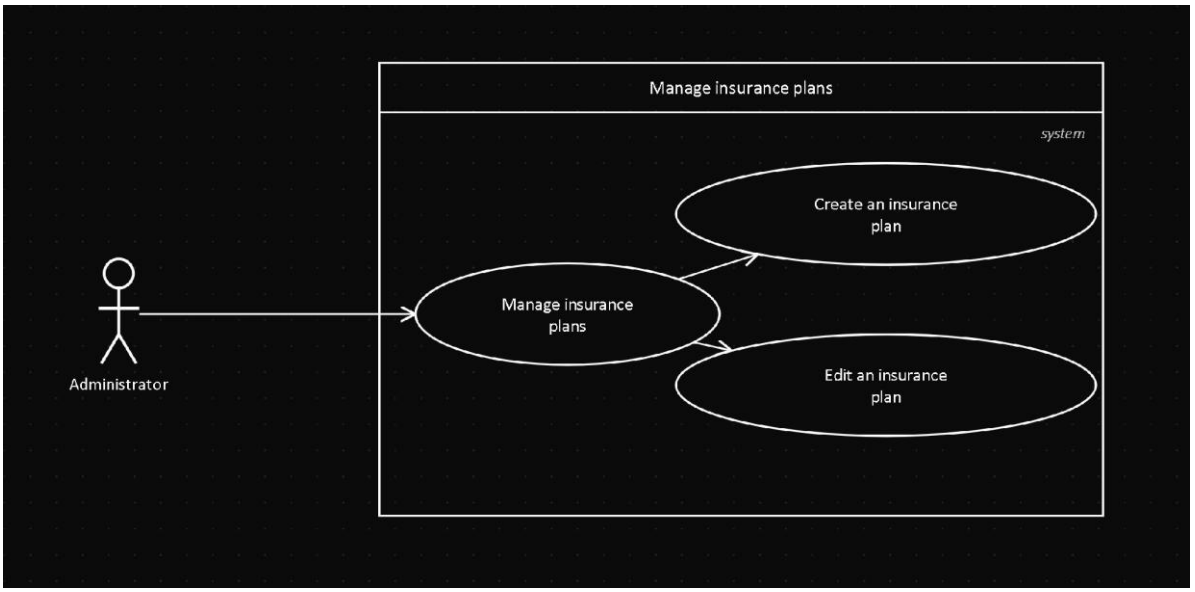
UC18 — Track vehicle location



Field	Value
Actor	Administrator
Related FR	FR6.1, FR6.2
Precondition	The vehicle is, or was, in `RENTED` status
Postcondition	The current location or the route history is shown

Description: the Administrator views the last known location of a rented vehicle, or the full route history of an already-finished rental.

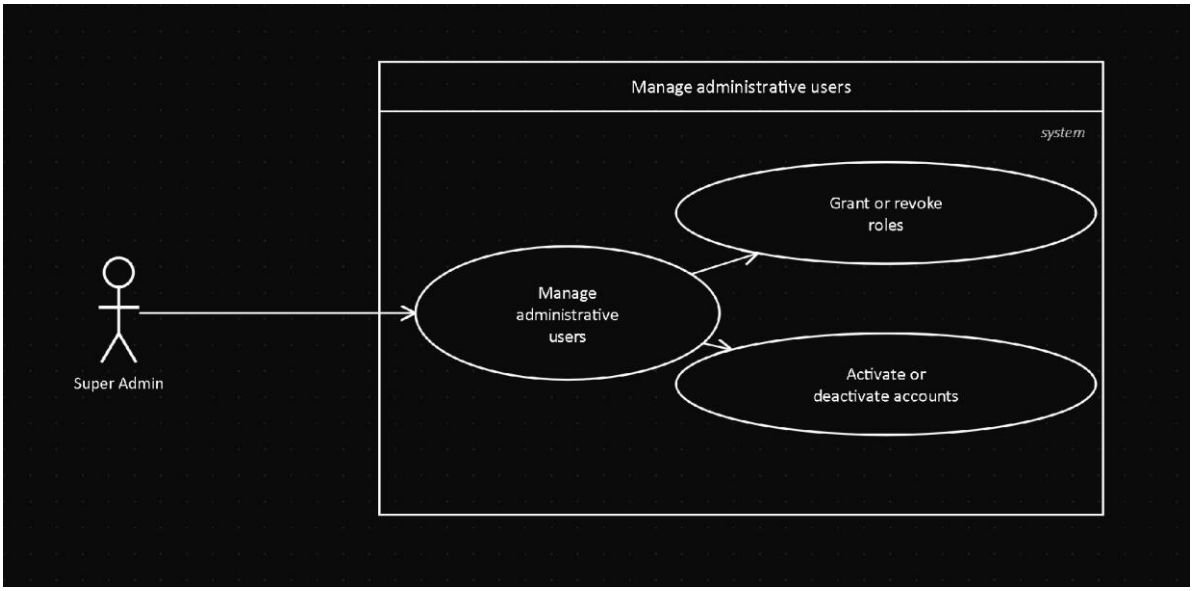
UC19 — Manage insurance plans



Field	Value
Actor	Administrator
Related FR	FR8.5
Precondition	The Administrator is authenticated
Postcondition	The insurance plan is created or edited

Description: the Administrator creates or edits the insurance plans (First Aid, Standard, All-Risk) that the Customer can choose when reserving.

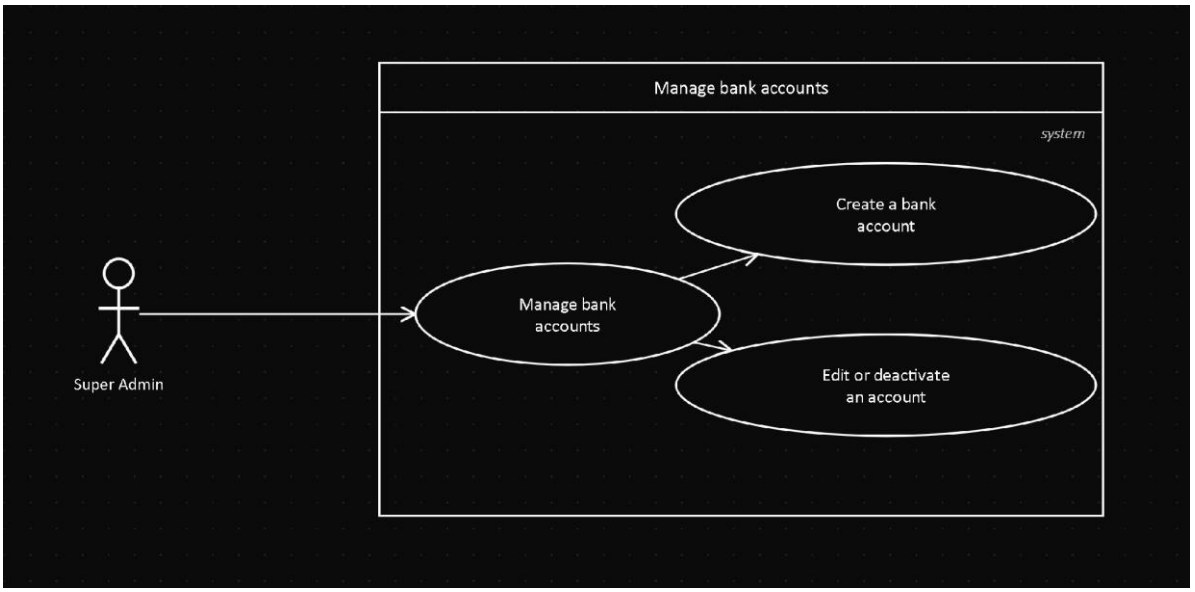
UC20 — Manage administrative users



Field	Value
Actor	Super Admin
Related FR	FR8.1
Precondition	The Super Admin is authenticated
Postcondition	The account's role or status is updated

Description: the Super Admin grants or revokes administrative roles, and activates or deactivates user accounts.

UC21 — Manage bank accounts



Field	Value
Actor	Super Admin
Related FR	FR8.6
Precondition	The Super Admin is authenticated
Postcondition	The bank account is created, edited or deactivated

Description: the Super Admin manages the company's bank accounts offered as a payment destination at checkout.

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5. References

Document title	Reference
Software Requirements Specification	`01-informe-especificacion-requisitos/SRS.md` (this repository)
Sequence diagrams	`02-Sequence-Diagrams.md` (this folder)
Activity diagrams	`03-Activity-Diagrams.md` (this folder)
State diagrams	`04-State-Diagrams.md` (this folder)